## 4.2.4 Analytical techniques

| | Learning outcomes | Additional guidance |
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| | *Learners should be able to demonstrate and apply their knowledge and understanding of:* | |
| **Infrared spectroscopy** | | |
| **(a)** | infrared (IR) radiation causes covalent bonds to vibrate more and absorb energy | |
| **(b)** | absorption of infrared radiation by atmospheric gases containing C=O, O–H and C–H bonds (e.g. $CO_2$, $H_2O$ and $CH_4$), the suspected link to global warming and resulting changes to energy usage | HSW9,10,11,12 Acceptance of scientific evidence explaining global warming has prompted governments towards policies to use renewable energy supplies. |
| **(c)** | use of an infrared spectrum of an organic compound to identify: | *M3.1* |
| | **(i)** an alcohol from an absorption peak of the O–H bond<br>(ii) an aldehyde or ketone from an absorption peak of the C=O bond<br>(iii) a carboxylic acid from an absorption peak of the C=O bond and a broad absorption peak of the O–H bond | In examinations, infrared absorption data will be provided on the *Data Sheet*.<br>Learners should be aware that most organic compounds produce a peak at approximately 3000 $cm^{-1}$ due to absorption by C–H bonds. |
| **(d)** | interpretations and predictions of an infrared spectrum of familiar or unfamiliar substances using supplied data | *M3.1*<br><br>Restricted to functional groups studied in this specification **(see also 6.3.2 e)**.<br><br>HSW3,5 Analysis and interpretation of spectra. |
| **(e)** | use of infrared spectroscopy to monitor gases causing air pollution (e.g. CO and NO from car emissions) and in modern breathalysers to measure ethanol in the breath | HSW12 Use of analytical techniques to provide evidence for law courts, e.g. drink driving. |
| **Mass spectrometry** | | |
| **(f)** | use of a mass spectrum of an organic compound to identify the molecular ion peak and hence to determine molecular mass | *M3.1*<br><br>Limited to ions with single charges.<br>Learners will **not** be expected to interpret mass spectra of organic halogen compounds.<br><br>Limited to organic compounds encountered in this specification **(see also 6.3.2 e)**.<br><br>Learners should be aware that mass spectra may contain a small M+1 peak from the small proportion of carbon-13.<br><br>HSW3,5 Analysis and interpretation of spectra. |

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| **(g)** analysis of fragmentation peaks in a mass spectrum to identify parts of structures. | *M3.1*<br><br>Learners should be able to suggest the structures of fragment ions.<br><br>HSW3,5 Analysis and interpretation of spectra. |

**Combined techniques**

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| **(h)** deduction of the structures of organic compounds from different analytical data including:<br><br>**(i)** elemental analysis **(see also 2.1.3c)**<br>**(ii)** mass spectra<br>**(iii)** IR spectra. | *M3.1*<br><br>Limited to functional groups encountered in this specification.<br><br>Learners will **not** be expected to interpret mass spectra of organic halogen compounds.<br><br>HSW3,5,6 Analysis and interpretation of different analytical data. |